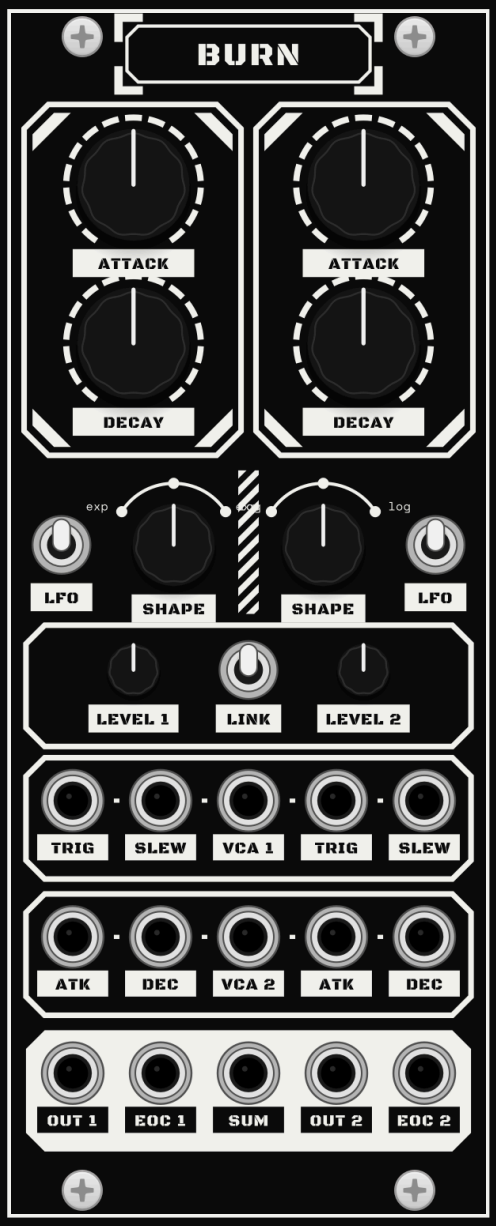
BETOBULLET

10 HP · EURORACK FORMAT
FOR VCV RACK 2



BURN

DUAL FUNCTION GENERATOR WITH VCAS

USER MANUAL · v2.0.0 · October 2026

CONTENTS

1. Introduction
2. Panel overview
3. Operation
4. Research data
5. Patch ideas
6. Specifications

Burn is a dual function generator with built-in VCAs: very snappy envelopes for percussion, ASR envelopes and slew from gates, LFOs from extremely slow to audio rate, end-of-cycle triggers, and the VCAs to make patching fast.

1. INTRODUCTION

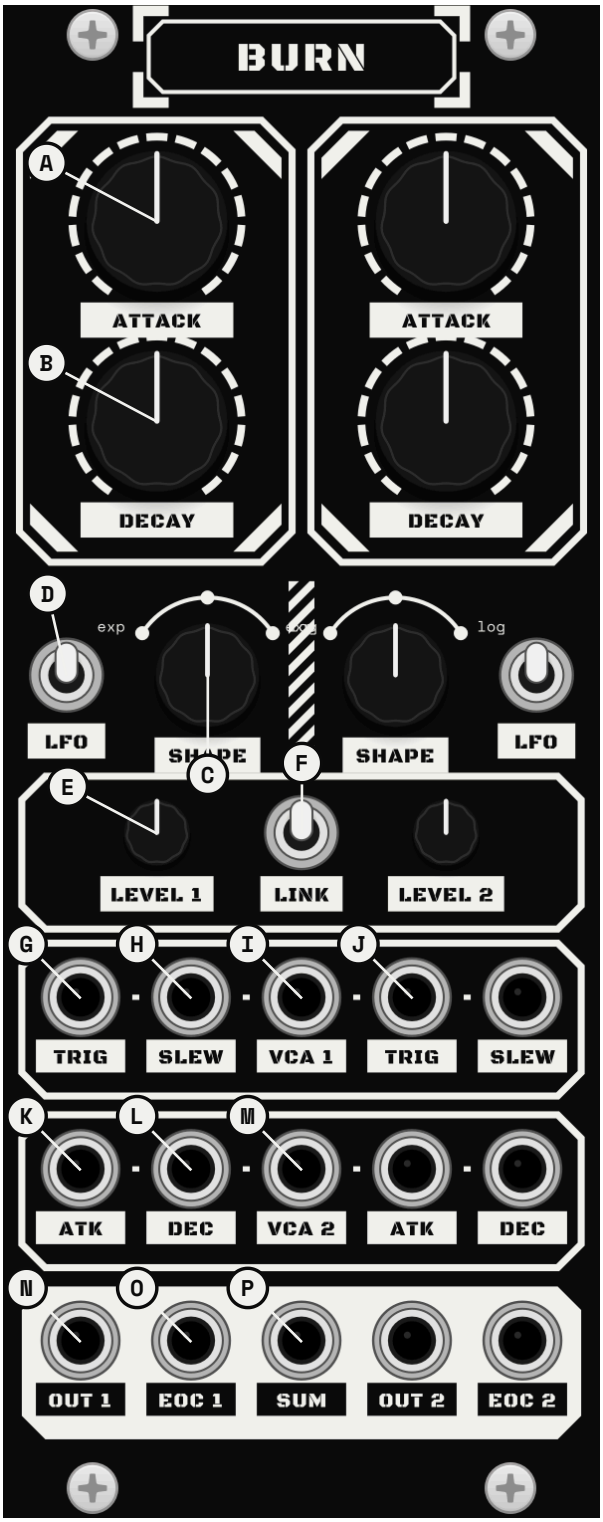
Burn packs two complete function generators into 10 HP. Each channel is a slew limiter driven by a pulse generator: a trigger produces an attack–decay envelope, a gate at the slew input produces an attack–sustain–release envelope, and any CV at the slew input comes out slewed. The LFO switch loops the envelope, from cycles of more than an hour to audio rate. A VCA on every channel shapes either the envelope itself or any signal you patch into it.

Burn recreates the published behaviour of the Cosmotronic Delta-V, an all-analogue module. The panel follows the lettering of the original manual (A–P).

INSTALLATION

Copy `Burn-2.0.0-win-x64.vcvplugin` to `Documents/Rack2/plugins` and restart VCV Rack. The module is listed under the brand **betobullet**.

2. PANEL OVERVIEW



A **Attack time**
Rise time, 0.3 ms to about 40 s. The tooltip shows the real time.

B **Decay time**
Fall time, 0.3 ms to about 40 s.

C **Slope shape**
Curve of both segments: exponential, linear or logarithmic.

D **One-shot / LFO switch**
In LFO mode the envelope retriggers itself at the end of every cycle.

E **Output attenuverter**
Level of the channel output, from −100 % to +100 % (±10 V).

F **Trigger link**
Connects trigger input G to channel 2 (G > J).

G **Trigger input**
Any rising signal fires an attack–decay envelope on channel 1.

H **Slew input**
A gate makes an ASR envelope; any CV comes out slewed.

I **VCA in, channel 1**
Normalled to 5 V: with nothing patched the output is the envelope.

J **Trigger input, channel 2**
Also fired by G when the link switch is on.

K **CV in for attack**
+1 V halves the attack time.

L **CV in for decay**
+1 V halves the decay time.

M **VCA in, channel 2**
Normalled to 5 V.

N **Channel 1 output**
Envelope × VCA × attenuverter.

O **End of cycle out**
A 0.5 ms trigger when the cycle has finished.

P **Sum out of 1 + 2**
Both channel outputs added.

3. OPERATION

The controls of channel 2 mirror those of channel 1 on the right half of the panel.

TRIGGER INPUT

The trigger input **(G)** takes any signal and turns its rising edge into a trigger pulse, generating an attack–decay envelope. **Envelopes ignore retriggers in the attack phase, but do retrigger in the decay phase**: a new trigger during the decay rises again from the current level. This allows clock and pulse divisions and musical rhythmic variations.

SLEW INPUT

The slew input **(H)** takes gate signals and slews them, but keeps a sustain as long as the gate is high. This creates a classic ASR envelope. Any other CV patched here comes out slew-limited, rising at the attack rate and falling at the decay rate.

VCA

The VCA inputs **(I, M)** have 5 V normalled to them. With nothing patched the output is the envelope, with the level set by the attenuverter **(E)**. Patch an audio source and the output is the enveloped audio. Patch a CV, for example from a sequencer, and you alter the amplitude of the envelope: great for accents and dynamics.

LFO SWITCH

The LFO switch **(D)** loops the envelope. It goes up to audio rate (about 1.6 kHz) or extremely slow: with negative CV at the attack and decay inputs, cycles of an hour are possible.

END OF CYCLE

Each channel has an end-of-cycle output **(O)** that sends a trigger after the cycle has finished, also on every loop in LFO mode.

TRIGGER LINK

The link switch **(F)** quickly connects both trigger inputs: trigger 1 then fires channel 2 as well, with no need for stackables or mults.

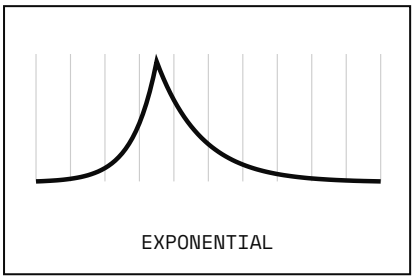
SUM OUTPUT

The sum output **(P)** adds both channel outputs. With opposite attenuverter settings it makes bipolar shapes.

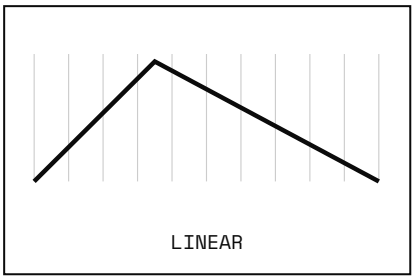
4. RESEARCH DATA

Responses measured on Burn (48 kHz, linear shape unless noted).

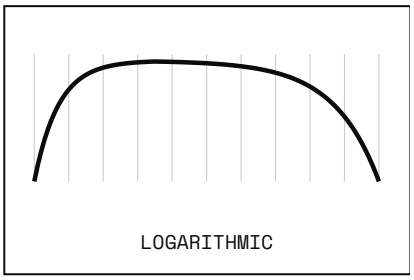
ENVELOPE SHAPES



EXPONENTIAL

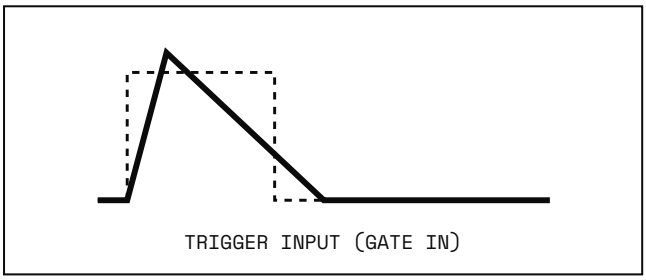


LINEAR

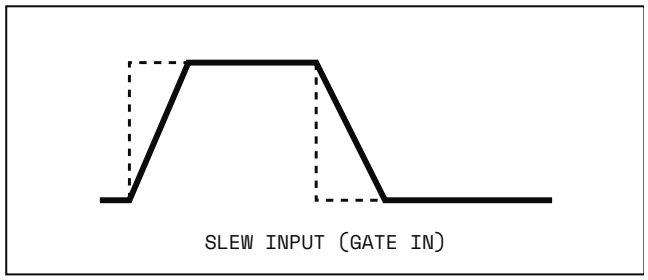


LOGARITHMIC

TRIGGER AND SLEW INPUTS



TRIGGER INPUT (GATE IN)



SLEW INPUT (GATE IN)

QUANTITY	VALUE
Max output	+10 V to −10 V, soft limited
Fastest rise / fall	0.3 ms each
Max LFO frequency	about 1.6 kHz
Slowest stage	about 40 s at the knob, over 20 min with −5 V of CV
EOC trigger width	0.5 ms
Time CV	+1 V halves the time (exponential)

5. PATCH IDEAS

- **Kick drum**: a clock into G, attack minimum, decay around 150 ms, shape exponential, a sine VCO into I.
- **Clock divider**: a long attack ignores the triggers that fall inside it. Take EOC as the divided clock.
- **Accented envelopes**: a stepped sequence into I sets the level of every hit.
- **Bouncing LFOs**: channel 1 in LFO mode; its EOC into J fires channel 2 once per cycle.
- **Portamento**: a pitch sequence into H, attack = decay, and take the output with E fully clockwise.

6. SPECIFICATIONS

ITEM	VALUE
Width	10 HP
Channels	2
Attack / decay	0.3 ms – ~40 s, CV: +1 V halves the time
Shape	exponential – linear – logarithmic, continuous
Modes	one-shot, LFO (looping)
VCA input	normalled to 5 V; output = envelope × VCA × 2 × level
Outputs	±10 V, EOC 10 V / 0.5 ms, sum
Retrigger	ignored during attack, honoured during decay

CREDITS

Design, DSP and panel: betobullet. Behaviour modelled on the Cosmotronic Delta-V user manual; Delta-V is a product of Cosmotronic and this is an independent recreation. Fonts: Black Ops One, Space Mono, Space Grotesk (SIL OFL).